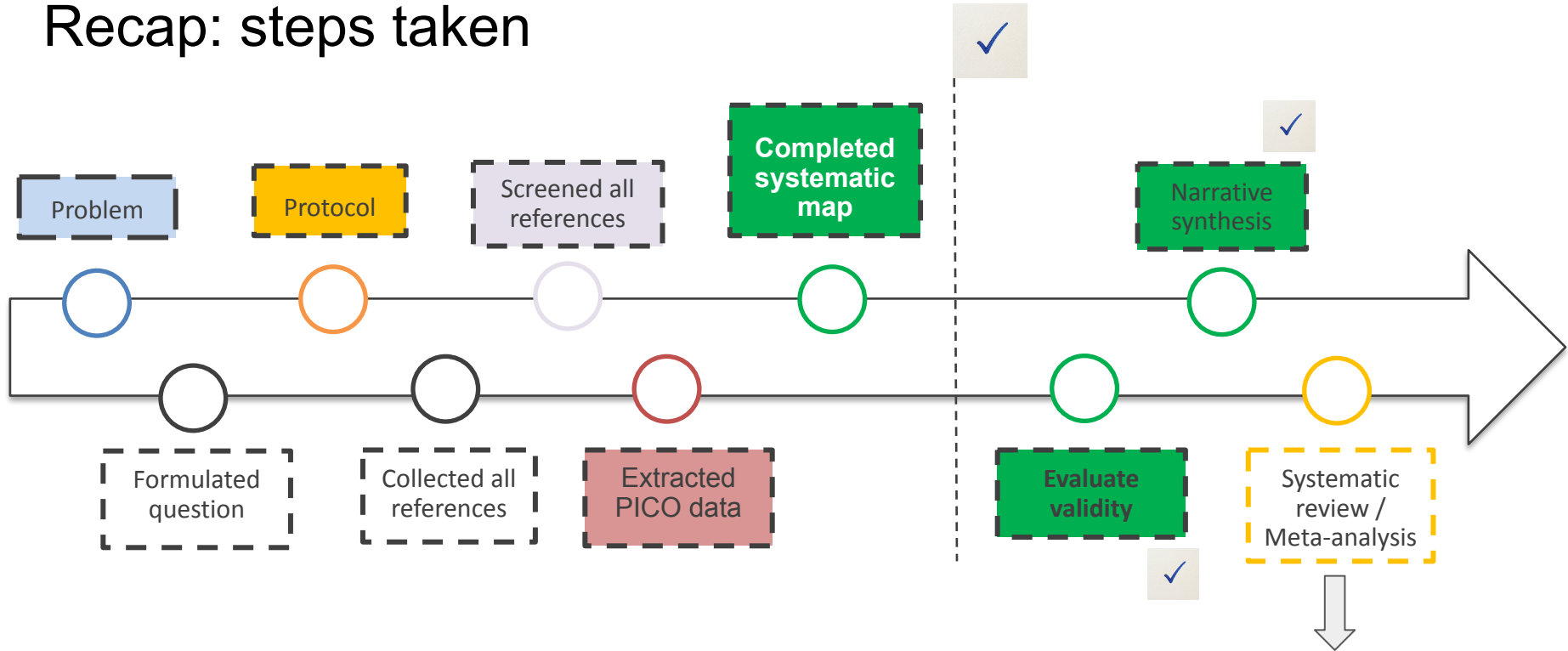


Quantitative data extraction

Importance of correctly formatting datasets

Recap: steps taken



But first, we need to extract quantitative data

Extracting data: what does it entail?

Extraction of the data is needed to calculate **effect sizes**

You need: **mean, sample size, sd/se/95% CI.**

+ extraction of variables that could explain the heterogeneity of effect sizes (***effect modifiers***)

Extract data from:

- Text
- Tables
- Figures
- supp. mats.
- Calculations may be needed (keep a record)



Extracting data: what does it entail?

! BE AWARE !

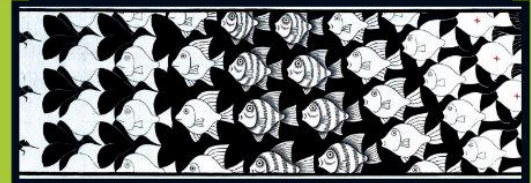
- Data extraction is **time-consuming**: clearly define the extraction sheet and the *effect modifiers* to be extracted.
- **Important to run pilot tests** using the drafted extraction sheet.
 - Do it on a small sample of articles to check that it **matches the content** of the studies.
- Decide what to do in case of **missing information** ("missing data", contact the authors, imputation)

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Recovering Missing or Partial Data from Studies:
A Survey of Conversions and Imputations
for Meta-analysis

Marc J. Lajeunesse

Handbook of Meta-analysis in Ecology and Evolution



EDITED BY

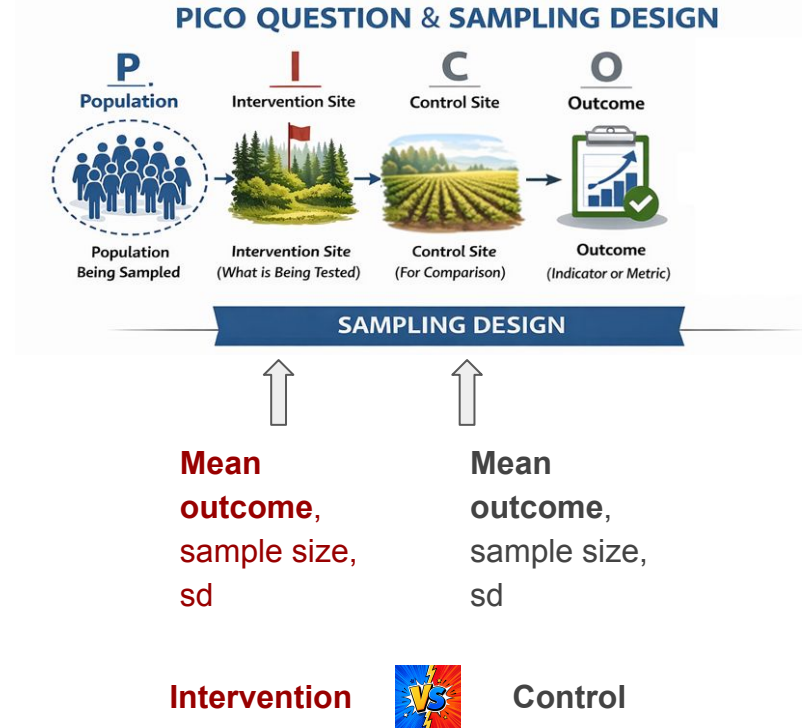
Julia Koricheva,
Jessica Gurevitch,
& Kerrie Mengersen

Format your data - (evidence)base - accordingly

What is a comparative study (follow your PICO)?

Example 1, Langridge et al. 2023:

- Population 1 × intervention 1 × richness
 - Population 1 × intervention 2 × abundance
- Same population (i.e. same taxonomic group), **different** interventions, and **different** measurements.
- **Two unique rows** in the database



Option 1: condensed database

condensed					
		sample size			
study	site	2000	2007	2010	measured outcomes
Hadley and Burns 20014	Shrewsbury	34	36		bird diversity, tree diversity, vegetation cover
Smith et al. 2012	Bangor			79	coleoptera body mass, tree diversity
Wagner et al. 2015	Leeds	12		15	bird diversity, tree diversity

1 line = 1 article:

- If **>1 measured outcome**? They are coded in the same cell.

Advantages :

- Easy to read.
- Easy to code

Disadvantages:

- If several outcomes: need to wrangle data for data analysis.
- **No link** can be made between *Sample size* et *Outcome* => « **linkage loss** »

Option 2: wide database

wide

study	site	sample.size.2000	sample.size.2007	sample.size.2010	outcome.bird.biodiversity	outcome.coleoptera.body.mass	outcome.tree.diversity	outcome.vegetation.cover
Hadley and Burns 20014	Shrewsbury	34	36		1		1	1
Smith et al. 2012	Bangor			79		1	1	
Wagner et al. 2015	Leeds	12		15	1		1	

1 line = 1 article:

- If >1 measured outcome? Outcomes are separated by columns (1 = yes : binary coding)

Advantages:

- Much easier to filter.
- Easier to synthesise.
- Columns can be combined/linked.

Disadvantages:

- **But again, we can lose** some information. In this example, **no link** can be made between *Sample size* et *Outcome* => « **linkage loss** »

Option 3: long database (best for meta-analysis)

long

study	site	year	sample.size	outcome
Hadley and Burns 20014	Shrewsbury	2000	34	bird biodiversity
Hadley and Burns 20014	Shrewsbury	2000	34	vegetation cover
Hadley and Burns 20014	Shrewsbury	2000	34	tree diversity
Hadley and Burns 20014	Shrewsbury	2007	36	bird biodiversity
Hadley and Burns 20014	Shrewsbury	2007	36	vegetation cover
Hadley and Burns 20014	Shrewsbury	2007	36	tree diversity
Smith et al. 2012	Bangor	2010	79	coleoptera body mass
Smith et al. 2012	Bangor	2010	79	tree diversity
Wagner et al. 2015	Leeds	2000	12	tree diversity
Wagner et al. 2015	Leeds	2000	12	bird diversity
Wagner et al. 2015	Leeds	2010	15	tree diversity
Wagner et al. 2015	Leeds	2010	15	bird diversity

1 line = 1 outcome:

- One line by comparative *study*.

Advantages :

- Ideal for filtering and synthesising.
- Can link sample sizes to outcomes (no linkage loss)
- **Best for meta-analyses**

Disadvantages:

- Difficult to read.
- The same article/study can be repeated several times (**i.e. separated into « study units »**).
- Need to devise a **study code** and **article code**

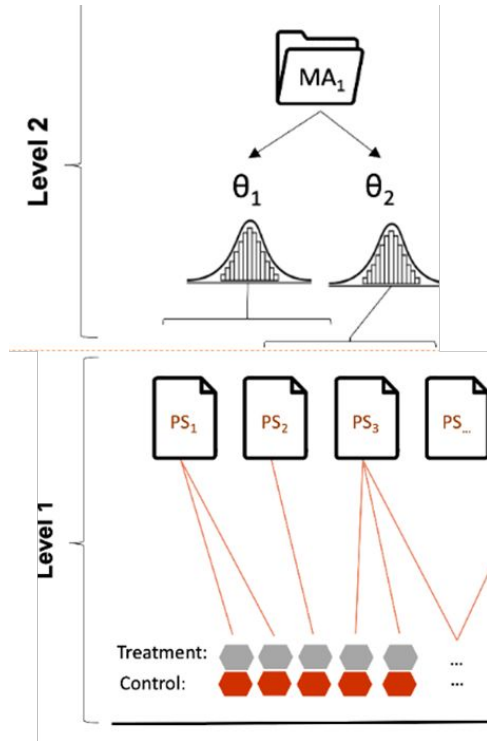
Consistency checking

Training test: data extraction of each study carried out **independently** by **2 people**.

Then,

- If several people share the workload, **check the consistency of data extraction** (formal pilot test) between people on a sample before starting the actual extraction and **discuss any disagreements**.
- If only 1 person, have someone to check a sample of the extracted data at the start of the work and **discuss any disagreements**.

Hierarchical data structure



Level 2: the meta-analysis: calculation of an overall average (i.e. aggregating the means of each *comparative study* within each individual *primary study*)

Level 1: individual *primary studies*



Comparative study or **paired data**: the individual primary studies report one or several paired data comparisons on an outcome of interest

Practical session and working example

Structure of extraction sheet (long)

Article Id	Biblio.ID	Biblio.Year	Comp.num	Comp.code	DIV.IND	M.DIV.UNM	SD.DIV.UNM	N.DIV.UNM	NBIS.DIV.UN	M.DIV.MAN	SD.DIV.MAN	N.DIV.MAN	NBIS.DIV.MA
28	28	2019	1	Orczewska_2019_1	SR	28.9	6.37	28	11	32.52	9.15	30	11
28	28	2019	2	Orczewska_2019_2	SR	4.63	2.37	28	11	9.78	5.16	30	11
28	28	2019	3	Orczewska_2019_3	SR	16.85	3.92	28	11	15.61	3.1	30	11
35	35	2019	1	Leski_2019_1	SR	11.41	6.33	4	1	20.79	5.82	4	1
35	35	2019	2	Leski_2019_2	SR	16.07	5.66	4	1	23.02	4.27	4	1
35	35	2019	3	Leski_2019_3	SR	26.84	6.09	4	1	26.59	8.31	4	1
35	35	2019	4	Leski_2019_4	SR	20.11	3.32	4	1	26.57	5.92	4	1
35	35	2019	5	Leski_2019_5	SR	32.45	2.82	4	1	32.43	3.4	4	1
35	35	2019	6	Leski_2019_6	SR	38.92	4.24	4	1	37.06	3.47	4	1
59	59	2019	1	Suominen_2019_1	SR	17.57	7.36	3	3	25.1	6.69	3	3
59	59	2019	2	Suominen_2019_2	SR	17.57	7.36	3	3	22.59	9.93	6	6

A Comparative study = a taxon group × an intervention × an outcome (SR, Abun)
 = e.g. lichens × stand not managed since at least 20 years × mean species richness

Comparative study level: ex. several possible comparators (specific extraction of timber)

Data extraction aligned with PICO elements

P I C O : population elements

Comparison code	Study Population: Kingdom	Study Population: branch	Study Population: Class	Study Population: Order	Study Population: Family	Taxonomic subset	Study Population: Ecological classification
Orczewska_2019_1	Plants	Vascular plants	NA	NA	NA	Total	Herbs
Orczewska_2019_2	Plants	Vascular plants	NA	NA	NA	Subset	Open-habitat species
Orczewska_2019_3	Plants	Vascular plants	NA	NA	NA	Subset	Ancient forest species

P I C O : comparator elements

MAN: crit.class	Management type	Management type 1	Management type 2	Management type 3
Clear cutting (cutting all the trees)	Salvage logging	1a	2aa	3b
Clear cutting (cutting all the trees)	Salvage logging	1a	2aa	3b
Clear cutting (cutting all the trees)	Salvage logging	1a	2aa	3b

Data extraction aligned with PICO elements

P I C O : **outcome** elements

Richness : Indice	MeanRichness: UnManaged	SD_SR : UnManaged	N_SR : UnManaged	MeanRichness: Managed	SD_SR : Managed	N_SR : Managed	Nbis_SR : Managed
SR	28,90	6,37	28	32,52	9,15	30	11
SR	4,63	2,37	28	9,78	5,16	30	11
SR	16,85	3,92	28	15,61	3,10	30	11

Top tip !
Code your data source

Data Comments
Data extract from boxplot FIG 4
Data extract from boxplot FIG 4
Data extract from boxplot FIG 4

<https://automeris.io/WebPlotDigitizer/>

<https://automeris.io/WebPlotDigitizer/>

Tools for extracting data: metaDigitise

Received: 13 July 2018 | Accepted: 12 October 2018

DOI: 10.1111/2041-210X.13118

APPLICATION

Methods in Ecology and Evolution 

Reproducible, flexible and high-throughput data extraction from primary literature: The METADIGITISE R package

Joel L. Pick  | Shinichi Nakagawa | Daniel W. A. Noble 



Advantages: possible to save, trace and modify data extraction

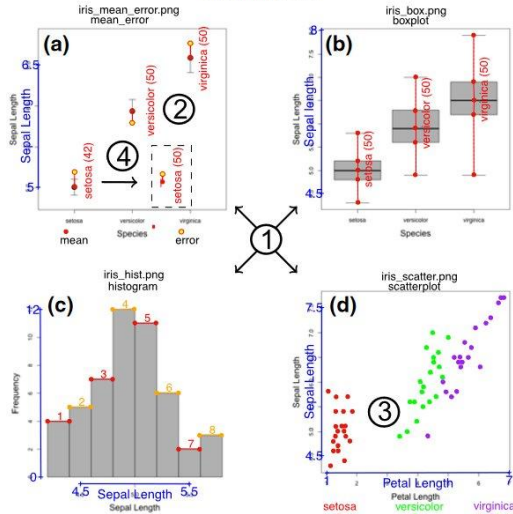
Disadvantages: no zoom

<https://cran.r-project.org/web/packages/metaDigitise/vignettes/metaDigitise.html>

Tools for extracting data: metaDigitise

metaDigitise

Plot interface



Data output

filename	variable	group_id	mean	sd	n	r	plot_type
iris_box.png	Sepal length	setosa	5.01	0.317	50	NA	boxplot
iris_box.png	Sepal length	versicolor	5.93	0.497	50	NA	boxplot
iris_box.png	Sepal length	virginica	6.49	0.603	50	NA	boxplot
iris_hist.png	Sepal Length	setosa	4.95	0.364	50	NA	histogram
iris_hist.png	Sepal Length	setosa	5.01	0.680	50	NA	mean_error
iris_hist.png	Sepal Length	versicolor	5.94	1.025	50	NA	mean_error
iris_hist.png	Sepal Length	virginica	6.59	1.251	50	NA	mean_error
iris_scatter.png	Petal Length	setosa	1.44	0.215	20	0.109	scatterplot
iris_scatter.png	Sepal Length	setosa	5.03	0.427	20	0.109	scatterplot
iris_scatter.png	Petal Length	versicolor	4.29	0.415	20	0.786	scatterplot
iris_scatter.png	Sepal Length	versicolor	5.97	0.603	20	0.786	scatterplot
iris_scatter.png	Petal Length	virginica	5.66	0.668	20	0.932	scatterplot

FUNCTIONALITY

① Different plot types

Capable of handling A) mean error plots, B) boxplots, C) histograms and D) scatterplots

② Entry of Metadata

Enter sample sizes variable and group names while digitising that are displayed on plot

③ Grouped Data

Enter as many groups as needed to capture descriptive statistics for sub-samples of data

④ Digitise, edit or replot digitisations

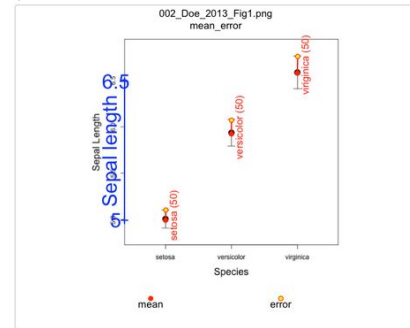
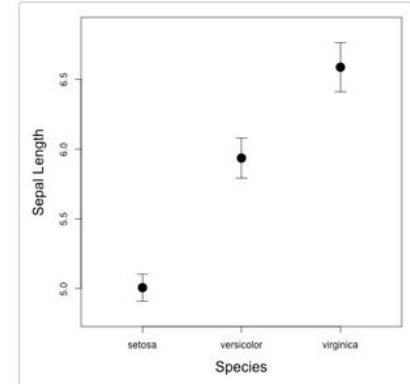
Simple user interface to guide user. Can digitise new images, edit digitisations or easily replot previous digitisations and metadata by cycling through images or choosing specific images

⑤ Summarising data

Get descriptive statistics automatically calculated for all plot types or use raw x,y data, if desired

⑥ Multiple image processing

Process as many images at once as needed and of varying types efficiently and quickly. New plots automatically plotted for digitisation



! Attention ! sometimes Rstudio buggs, use R 'base'

Practical session

Question: estimate the effect of abandonment of forest management on species richness of multiple taxons

Sample of 3 articles:

A **comparative study** = a taxon × an intervention × an outcome

Practical session

Extracting data

1. Install et load metaDigitise package; set working directory
2. Prepare the figure files (screen capture .png), put them in a "figs" folder in the working directory
3. Run `metadigitise > dat <- metaDigitise(dir = "./figs")`
4. Use package to extract data into an Excel file
5. If you need to convert a se to sd cf. excel f $SD = SE \times \sqrt{n}$

Discussion

```
install.packages("metaDigitise")
```

```
library(metaDigitise)
```

```
install.packages("writexl")
```

```
library(writexl)
```

```
extract_dat <- metaDigitise("C:/Users/33676/Documents/blahblahala/Metadigitise")
```

```
write_xlsx(extract_dat, "metadigitise_output.xlsx")
```

